

Figure 1.3. Structure of Sphingolipids.

In sphingolipids, the hydrophobic region consists of a long chain sphingoid base with generally 18 carbons, which is linked to the acyl group of a fatty acid via an amide bond. The polar head group (X) attached to C1 consists of X=H in the parent compound, ceramide.

In biological membrane, the sphingolipid components that contain a sugar molecule are referred to as glycolipids. These lipids obtain the name from sphingosine with a polar head group attached to C1 (Figure 1.3). Sphingolipids are abundantly present in membrane. There are various sphingolipids depending on (i) the fatty acid attached to C2 by an amide bond, or (ii) nature of polar head group esterified to the hydroxyl group at C1. Sphingolipids with one or more sugars are called glycophospholipids, which may be either neutral or negatively charged. In some case, a phosphate ester can link a base (choline or ethanolamine similar to phosphoglycerides) to C1 of sphingolipids and it is called sphingomyelin. Complex glycolipids like gangliosides contain oligosaccharides as their polar head group and one or more residues of sialic acid as the terminal sugar.

Sterols

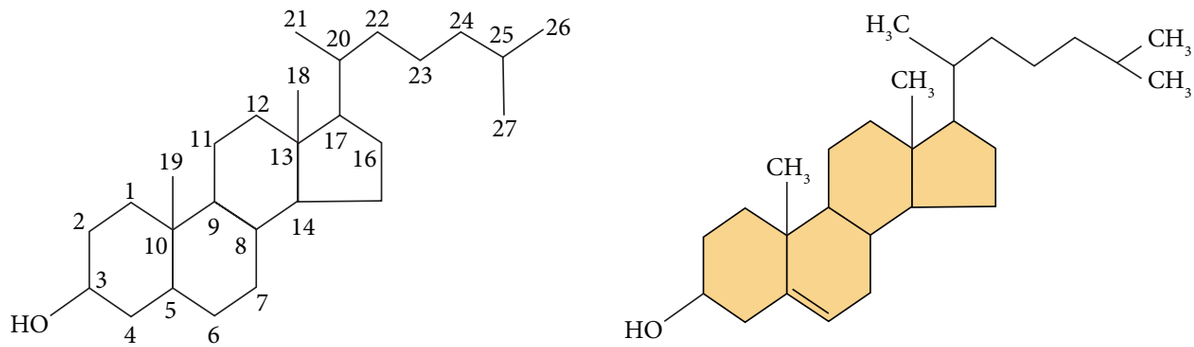


Figure 1.4: Structure of sterols

Sterols are third major lipid constituents of the lipid bilayer (Figure 1.4) cholesterol is the major type of sterol found in animal membranes while flat sterols can be found in bacteria, lower eukaryotes and plants. The cyclopentanoperhydrophenanthrene ring of cholesterol is apolar, inserting into the core of the lipid bilayer with the hydroxyl group at C3 situated at the surface. The precursor moieties of cholesterol namely isoprenol, geranyl and farnesylisoprenols act as hydrocarbon anchors for various membrane proteins.

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